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about_Functions_Advanced

Short description

Introduces advanced functions that are a way to create cmdlets using scripts.

Long description

A cmdlet is a single command that participates in the pipeline semantics of PowerShell. This includes binary cmdlets, PowerShell advanced functions, and CDXML cmdlets.

Advanced functions allow you create cmdlets that are written as a PowerShell function. Advanced functions make it easier to create cmdlets without having to write and compile a binary cmdlet. Binary cmdlets are .NET classes that are written in a .NET language such as C#.

Advanced functions use the `CmdletBinding` attribute to identify them as functions that act like cmdlets. The `CmdletBinding` attribute is similar to the Cmdlet attribute that's used in compiled cmdlet classes to identify the class as a cmdlet. For more information about this attribute, see about_Functions_CmdletBindingAttribute.

The parameters of the function are variables declared in the `param()` statement. You can use the optional `[Parameter()]` attribute alone or in combination with the `[Alias()]` attribute or any of the parameter validation attributes. For more information about how to declare parameters, see about_Functions_Advanced_Parameters.

The following example shows a function that accepts a name and then prints a greeting using the supplied name. Also notice that this function defines a name that includes a verb (Send) and noun (Greeting) pair like the verb-noun pair of a compiled cmdlet. However, functions aren't required to have a verb-noun name.

```
function Send-Greeting
{
    [CmdletBinding()]
    param(
        [Parameter(Mandatory=$true)]
        [string]$Name
    )
```

```
    process
    {
        Write-Host ("Hello " + $Name + "!")
    }
}
```

This function performs the work in the `process` block, which is equivalent to the **ProcessingRecord** method used in compiled cmdlets. The `process` block and the other named blocks are described in about_Functions_Advanced_Methods.

Advanced functions differ from compiled cmdlets in the following ways:

- Advanced function parameter binding doesn't throw an exception when an array of strings is bound to a **Boolean** parameter.
- The `ValidateSet` attribute and the `ValidatePattern` attribute can't pass named parameters.
- Advanced functions can't be used in transactions.

PIPELINESTOPTOKEN

Beginning with PowerShell 7.6-preview.4, `$PSCmdlet` includes the `PipelineStopToken` property allowing access a CancellationToken tied to the PowerShell stop event source. The token is triggered when the PowerShell pipeline receives a request to stop. Use it with a .NET method that accepts a `CancellationToken` overload to exit the method when requested rather than waiting until the method returns.

In the following example, the function is calling `HttpClient.GetStringAsync`, which can take time to respond when the network is slow or there is a lot of data being returned.

```
function Invoke-CancelableWebRequest {
    [CmdletBinding()]
    param(
        [Parameter(Mandatory=$true)]
        [string]
        $Url
    )

    $client = [System.Net.Http.HttpClient]::new()
    $client.GetStringAsync(
        $Url,
        $PSCmdlet.PipelineStopToken).GetAwaiter().GetResult()
}

Invoke-CancelableWebRequest -Url https://httpbin.org/delay/10
# Press ctrl+c to cancel
```

See also

- about_Functions
- about_Functions_Advanced_Methods
- about_Functions_Advanced_Parameters
- about_Functions_CmdletBindingAttribute
- about_Functions_OutputTypeAttribute

- about_Variables